

CS23331-Design and Analysis of Algorithms

Problem 1: Finding Complexity using Counter Method

Completion requirements

Convert the following algorithm into a program and find its time complexity using the counter method.

void function (int n)

```
{
    int i= 1;
    int s =1;
    while(s <= n)
    {
        i++;
        s += i;
    }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Answer:

```
#include<stdio.h>
```

```
int main(){
```

```
    int n;
```

```
    scanf("%d",&n);
```

```
    int i=1;
```

```
    int s=1;
```

```
    int count=2;
```

```
    while(s<=n){
```

```

    count++;

    i++;

    count++;

    s+=i;

    count++;

}

int answer=count+1;

printf("%d",answer);

return 0;

}

```

Input	Expected	Got	
	9	12	12
	4	9	9

Problem 2: Finding Complexity using Counter method

Convert the following algorithm into a program and find its time complexity using the counter method.

```

void func(int n)
{
    if(n==1)
    {
        printf("*");
    }
    else
    {
        for(int i=1; i<=n; i++)
        {
            for(int j=1; j<=n; j++)
            {
                printf("*");
                printf("*");
            }
        }
    }
}

```

```

        break;
    }
}
}

```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
#include<stdio.h>
```

```
int main(){
```

```
    int n;
```

```
    scanf("%d",&n);
```

```
    int count=0;
```

```
    count++;
```

```
    if(n==1)
```

```
    {
```

```
        printf("*");
```

```
    }
```

```
    else
```

```
    {
```

```
        for(int i=1; i<=n; i++)
```

```
        {
```

```
            count++;
```

```
            for(int j=1; j<=n; j++)
```

```
            {
```

```
                count++;
```

```
                count++;
```

```
                count++;
```

```
                break;
```

```
            }
```

```
        count++;;
```

```

    }

    count++;

}

printf("%d",count);

return 0;

}

```

Input	Expected	Got	
	2	12	12
	1000	5002	5002
	143	717	717

Problem 3: Finding Complexity using Counter Method

Convert the following algorithm into a program and find its time complexity using counter method.

```

Factor(num) {
{
    for (i = 1; i <= num;++i)
    {
        if (num % i== 0)
        {
            printf("%d ", i);
        }
    }
}
}

```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
#include <stdio.h>
```

```
int main()
```

```

{
    int num;
    scanf("%d",&num);
    int count = 0;
    int i;
    for(i = 1; i <= num; i++)
    {
        count++;

        if(num % i == 0)
        {
            count++;
        }
        count++;
    }
    count++;
    printf("%d",count);

    return 0;
}

```

Input	Expected	Got	
	12	31	31
	25	54	54
	4	12	12

Problem 4: Finding Complexity using Counter Method

Convert the following algorithm into a program and find its time complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
#include<stdio.h>
```

```
int main(){
```

```
    int n;
```

```
    scanf("%d",&n);
```

```
    int c=0;
```

```
    int count=2;
```

```
    for(int i=n/2;i<n;i++){
```

```
        count++;
```

```
        for(int j=1;j<n;j=2*j){
```

```
            count++;
```

```
            for(int k=1;k<n;k=k*2){
```

```
                count++;
```

```
                c++;
```

```
                count++;
```

```
            }
```

```
            count++;
```

```
        }
```

```
        count++;
```

```
    }
```

```
    printf("%d",count);
```

```

return 0;
}

```

Input	Expected	Got	
	4	30	30
	10	212	212

Problem 5: Finding Complexity using counter method

Convert the following algorithm into a program and find its time complexity using counter method.

```

void reverse(int n)
{
    int rev = 0, remainder;
    while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n/= 10;
    }
    print(rev);
}

```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```

#include<stdio.h>
int main(){
    int n;
    scanf("%d",&n);
    int rev=0,rem;
    int count=0;
    while(n!=0){
        count++;
        rem=n%10;
        count++;
        rev=rev*10+rem;
        count++;
        n=n/10;
    }
}

```

```
        count++;  
    }  
    count+=3;  
    printf("%d",count);  
    return 0;  
}
```

Input	Expected	Got	
	12	11	11
	1234	19	19